Roll No.: _

Amrita Vishwa Vidyapeetham, Coimbatore Campus

B.Tech Mid-Term Examinations – August 2026

Fifth Semester

Computer Science and Engineering

23CSE304 Embedded Systems

Duration: Two hours Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Understand the fundamentals, functions, and architecture of embedded systems.
CO02	Develop applications making use of GPIOs and peripheral units using Embedded C.
CO03	Develop applications for the usage of Timers, Interrupts, and Debugging of embedded applications.
CO04	Design an embedded system using an ARM processor.

Answer all questions

PART A (Theory)[20 Marks]

1) A washing machine automatically controls water intake, washing time, motor operation, and completion of the washing cycle based on the selected program. Explain why the washing machine is considered an embedded system and describe the characteristics of embedded systems reflected in this application. [6] [CO01][BTL 2]

2) **Answer the following**
   a. Differentiate between Thread mode and Handler mode in ARM Cortex-M processors. Which of these modes is used exclusively for privileged operations?
   b. Can the system switch from unprivileged mode to privileged mode using software? Explain briefly. [3] [CO01][BTL 2]

3) State one function of the SysTick timer in ARM Cortex-M4. Calculate the reload value to be loaded into the SysTick Reload Value Register (SYST_RVR) to generate a periodic interrupt every 1 ms, assuming a processor clock of 84 MHz with a /8 prescaler.
[3] [CO03][BTL 2]

4)
   a. Apply the concept of bit-banding in ARM Cortex-M4 to control a GPIO pin using STM32F401CCU6. The GPIOA base address is 0x40020000 and the GPIO output data register (ODR) offset is 0x14.

R

i. Determine the address of the GPIOA_ODR register and calculate the bit-band alias address corresponding to bit 7 of the register.
ii. Explain how bit-banding improves GPIO control in real-time applications

[4] [CO02][BTL 3]

b. You are working with a 32-bit control register in the STM32F401CCU6 microcontroller. The register enables and disables multiple hardware features using different bits, and several bits are already set.
   i. Analyze the bitwise operation required to clear bit 7 while preserving the states of all other bits in the register.
   ii. Compare this approach with the use of a direct assignment such as REG = 0;.

[4] [CO02][BTL 4]

PART B (Practical)[30 Marks]

5) Design and implement a **Smart Lighting with Door Sensor and Manual Override system** on an STM32F401CCU6 microcontroller. The hardware setup is as follows: switches are connected to PA0 (door sensor) and PB10 (manual override), an LDR is connected to channel 1 through the ADC, and LEDs are connected to PC13 and PB9. The system should work as follows:
   a. when the switch on PA0 is pressed, it simulates a door opening and LED PC13 must turn ON, while releasing the switch (door closed) should turn it OFF.
   b. The LDR on PA1 is used for automatic lighting: when the ambient light is low (ADC value - set threshold), LED PB9 should turn ON, and when the light is high, it should turn OFF.
   c. In addition, the switch on PB10 acts as a manual override for LED PB9; pressing it should temporarily disable the auto-light function and toggle the LED (PB9) ON or OFF with each press, resuming automatic LDR control once override is released.

Rubrics:

Register-Level Configuration	[5] [CO02][BTL 2]
Door Sensor Functionality (PA0 → PC13)	[5] [CO02][BTL 3]
LDR-based Automatic Lighting (PA1 ADC → PB9)	[5] [CO02][BTL 3]
Manual Override Function (PB10 → PB9)	[5] [CO02][BTL 3]
ADC Value Demonstration & Functional Verification	[10] [CO02][BTL 4]

****

R

Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**9**	BTL 1	
CO02	**38**	BTL 2	**17**
CO03	**3**	BTL 3	**19**
CO04		BTL 4	**14**
CO05		BTL 5	
CO06		BTL 6	

Bloom's Taxonomy Levels (attached for reference)

Level 1 – Remember

Level 2 - Understand

Level 3 – Apply

Level 4 - Analyze

Level 5 - Evaluate

Level 6- Create

R